

Instruction manual for GreenMean Project

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1 For Survey Participant

1.1 Go to GreenMean project site

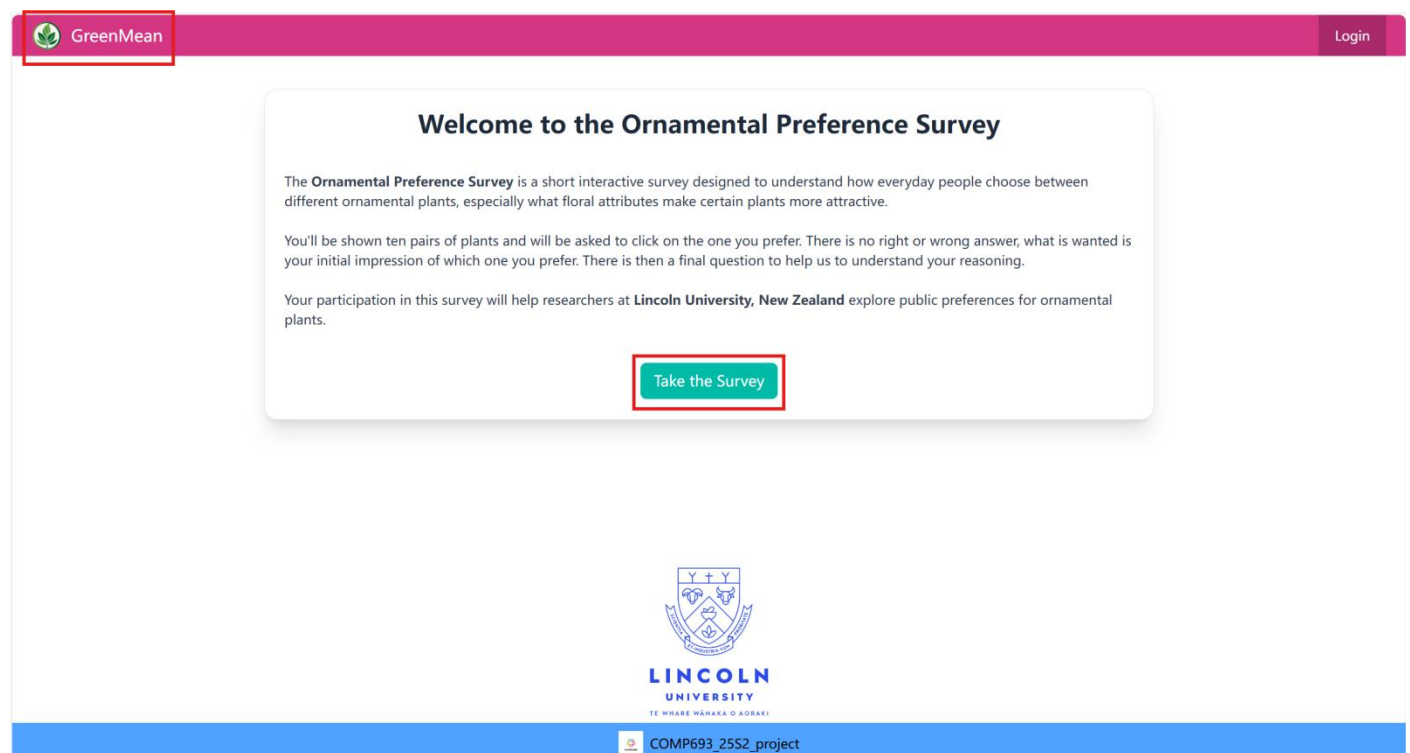
Click the link below, or scan the barcode on a mobile phone

<https://greenmean.pythonanywhere.com/home/>

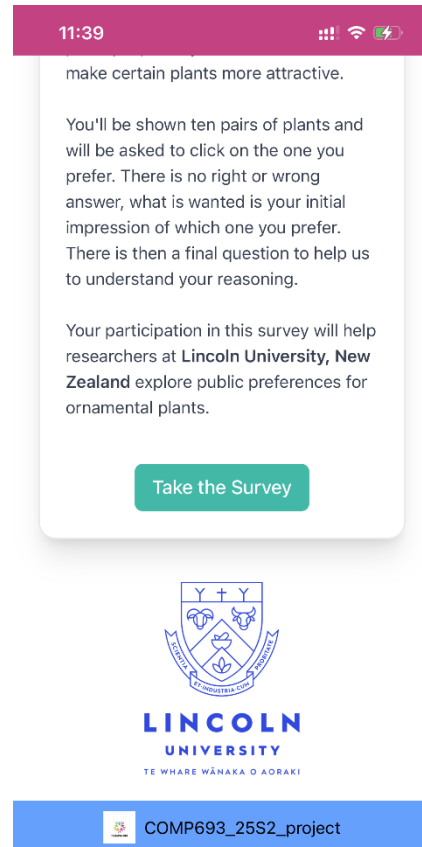
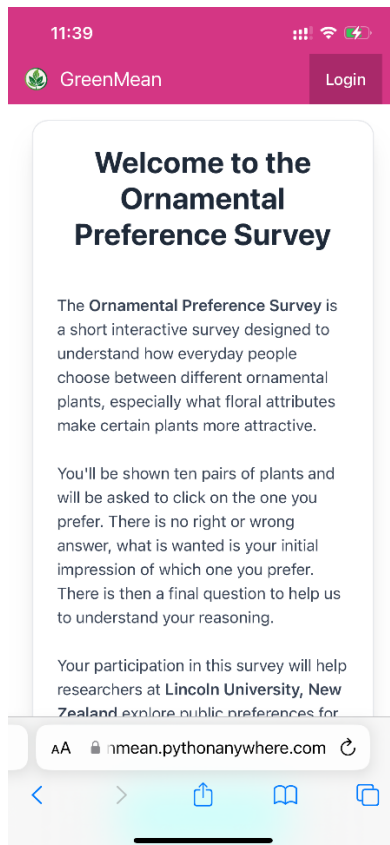


1.2 Click the Take the Survey button to start

If the button is not available, then click the logo on top left.

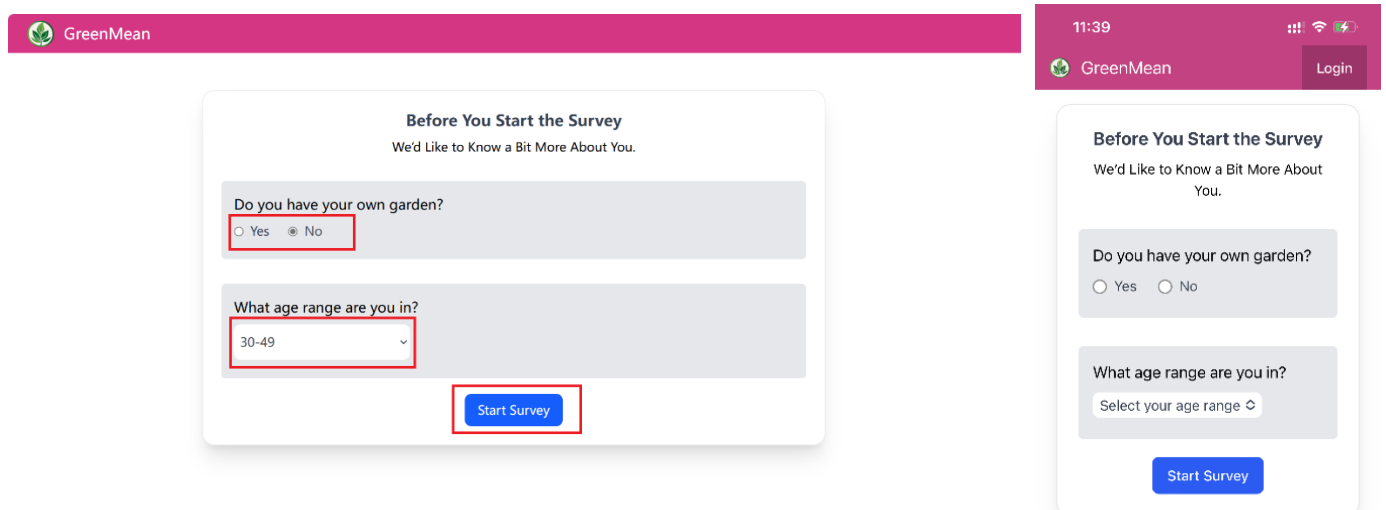


Mobile version



1.2 Complete the two pre-questions

Then click the **Start Survey** button



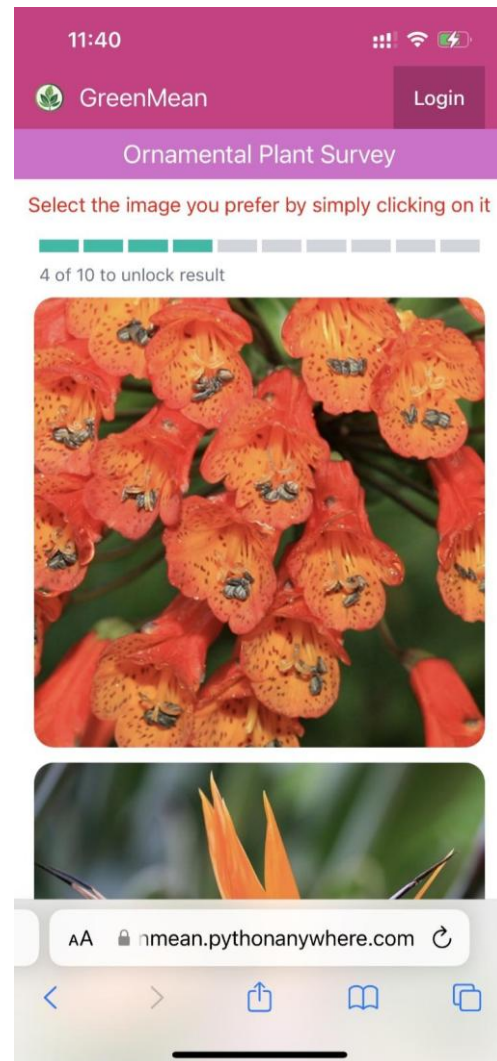
1.3 Start survey by click the image the is preferred

There are 10 pairs of images in total, and a progress bar is displayed at the top.

Select the image you prefer by simply clicking on it



Mobile version:



1.4 Choose an option for the final preference question.

GreenMean

Login

Final Question

What characteristics most influenced your choice of plant?

☐ The dominant flower colour (e.g. red, blue, white etc.)

☒ Contrasting colours in the flower (e.g. green and yellow, blue and white etc.)

☐ Flower shape (e.g. round, heart-shaped etc.)

☐ It was a plant that was familiar to you

☐ It was a plant that looked different from most plants you know

Submit

11:41



Final Question

What characteristics most influenced your choice of plant?

☐ The dominant flower colour (e.g. red, blue, white etc.)

☐ Contrasting colours in the flower (e.g. green and yellow, blue and white etc.)

☐ Flower shape (e.g. round, heart-shaped etc.)

☐ It was a plant that was familiar to you

☐ It was a plant that looked different from most plants you know

Submit

1.5 View survey summary and welcome message

Click the **Your Choices** button to view selection detail

The image displays two versions of a mobile application interface side-by-side. The left version is a desktop mockup, and the right version is a mobile phone screen mockup. Both screens show a 'Thank You for taking part in our survey' message, a paragraph about the survey, a 'Your Selection Summary' section, and two boxes showing 'Invasive Plants Chosen' (80.0%) and 'Non-Invasive Plants Chosen' (20.0%). The mobile version has a smaller font and a different layout for the summary section.

Desktop Version (Left):

- Header: GreenMean
- Text: Thank You for taking part in our survey
- Text: In each pair of images, you were offered an ornamental plant that is known to be an environmental weed (invasive plant) and one that is thought to be safe.
- Text: Your preferences highlight whether you are more likely to choose one type of plant over another. The higher your selection of plants that are safe, the better for the environment.
- Section: Your Selection Summary
- Box 1: Invasive Plants Chosen, 80.0%
- Box 2: Non-Invasive Plants Chosen, 20.0%
- Buttons: Back to Home, Your Choices

Mobile Version (Right):

- Header: 11:42, Signal, Wi-Fi, Battery
- Text: Your preferences highlight whether you are more likely to choose one type of plant over another. The higher your selection of plants that are safe, the better for the environment.
- Section: Your Selection Summary
- Box 1: Invasive Plants Chosen, 70.0%
- Box 2: Non-Invasive Plants Chosen, 30.0%
- Buttons: Back to Home, Your Choices

1.6 Choice detail page

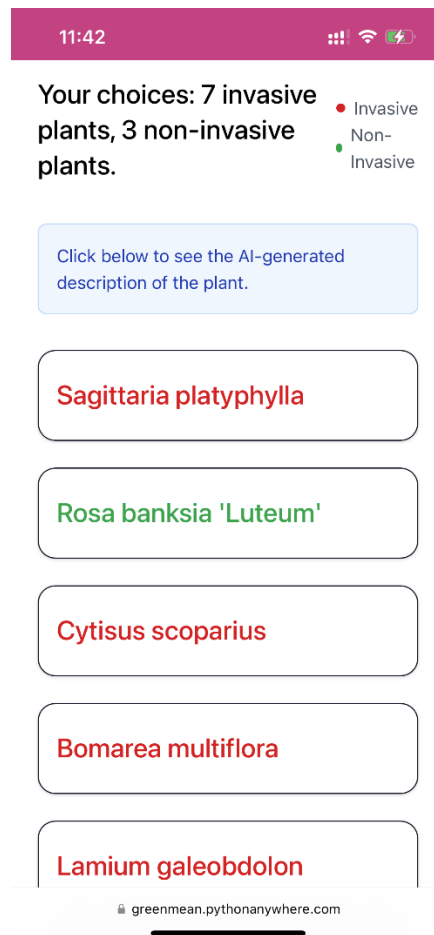
View the selected plant names that was chosen. Invasive plants are red, and non-invasive plants are green.

GreenMean

Login

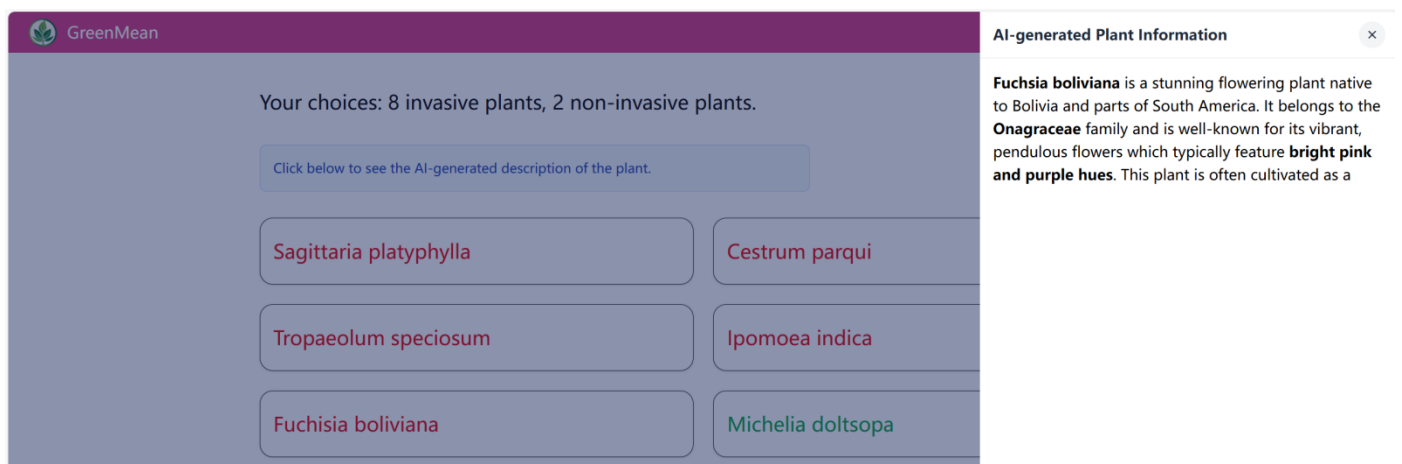
Your choices: 8 invasive plants, 2 non-invasive plants.

</

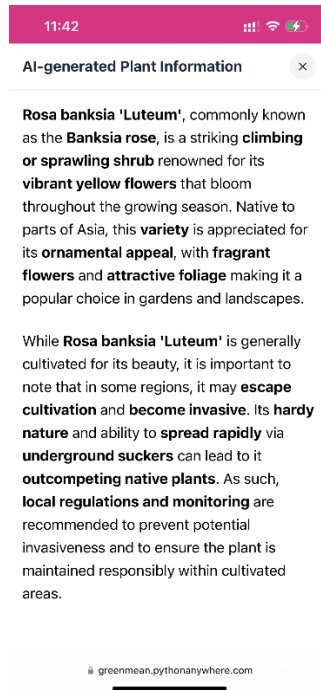


1.7 View AI Generated introduction for that plant

Click on a plant name to view a short introduction (2-3 paragraphs) about that plant. The information will appear on the right side in an off-canvas panel.



Mobile version:



2 For site administrator

2.1 Login

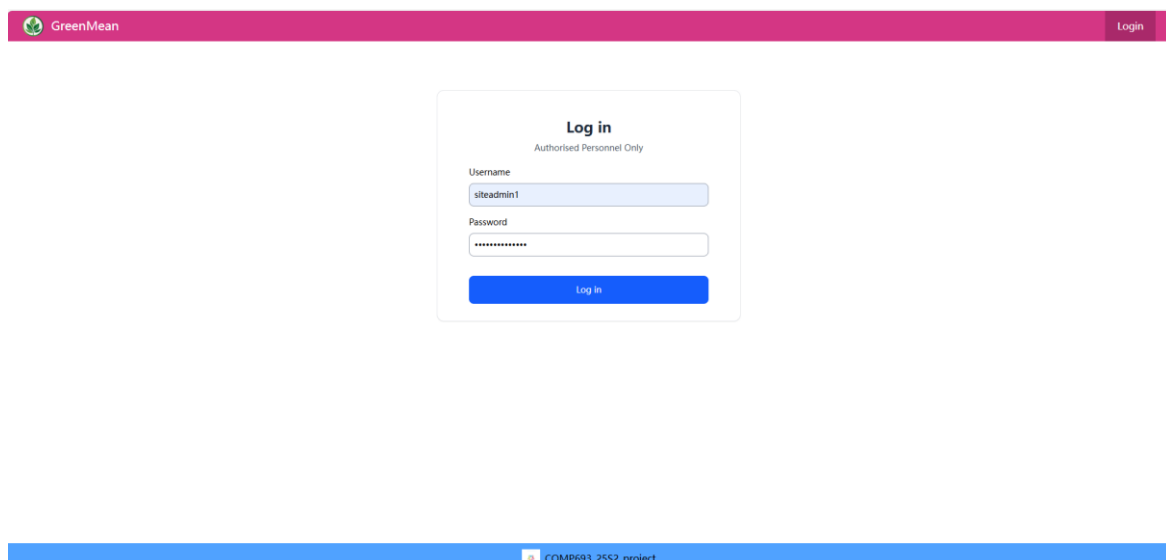
Use following credentials to login the application. For a better user experience, access the admin site on a desktop, as it contains dashboards with charts and tables.

Username: siteadmin1

Password: siteadmin1pass

Username: siteadmin2

Password: siteadmin2pass



2.2 Dashboard

Click the dashboard menu

GreenMean

DashboardSite Management

Welcome to the Ornamental Preference Survey

The **Ornamental Preference Survey** is a short interactive survey designed to understand how everyday people choose between different ornamental plants, especially what floral attributes make certain plants more attractive.

You'll be shown ten pairs of plants and will be asked to click on the one you prefer. There is no right or wrong answer, what is wanted is your initial impression of which one you prefer. There is then a final question to help us to understand your reasoning.

Your participation in this survey will help researchers at **Lincoln University, New Zealand** explore public preferences for ornamental plants.

The page shows both invasive and non-invasive plant choices along with the accumulated percentage and count of all survey results.

At the top, a bar chart and summary numbers are displayed. Below, the raw survey data is shown and can be downloaded.

GreenMean

DashboardSite Management

OverallBy Age GroupBy GardeningBy PreferenceBradley-Terry Model

Analysis of Plant Invasiveness vs. Attractiveness

Overall proportion of choices

Invasive Choices

59.11%146

Non-Invasive Choices

40.89%101

Category	Percentage	Count
Invasive	59.11%	146
Non-Invasive	40.89%	101

Download Data

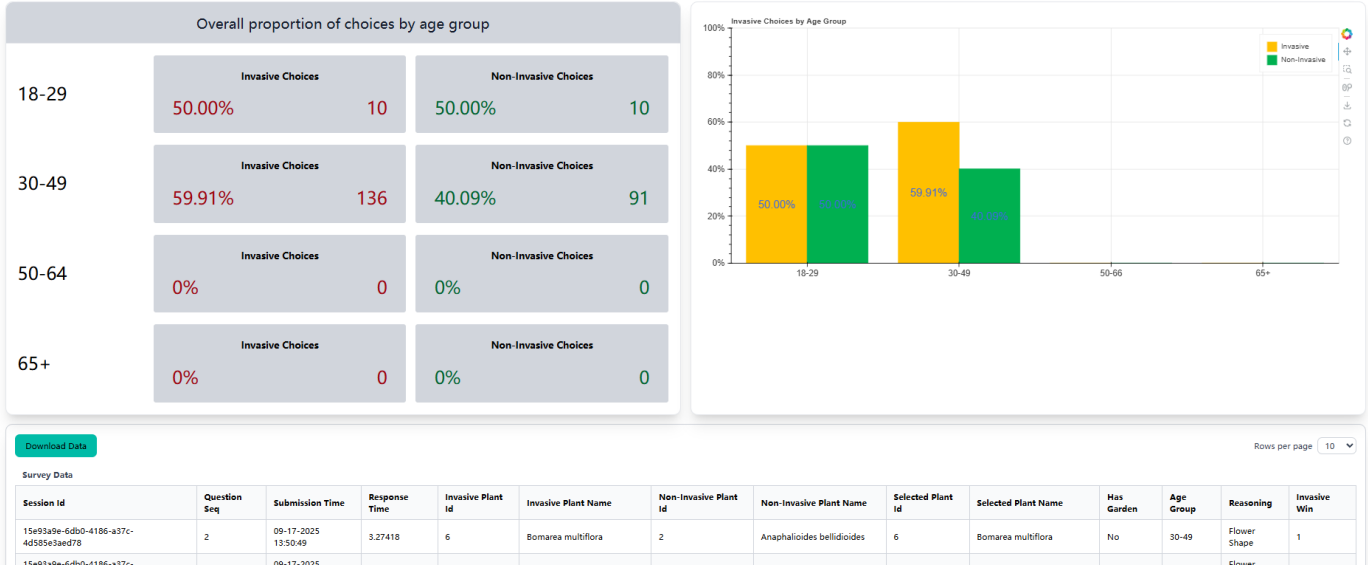
Rows per page 10

Session Id	Question Seq	Submission Time	Response Time	Invasive Plant Id	Invasive Plant Name	Non-Invasive Plant Id	Non-Invasive Plant Name	Selected Plant Id	Selected Plant Name	Has Garden	Age Group	Reasoning	Invasive Win
15e93a9e-6db0-4186-a37c-4d585e3aed78	2	09-17-2025 13:50:49	3.27418	6	Bomarea multiflora	2	Anaphalioides bellidioides	6	Bomarea multiflora	No	30-49	Flower Shape	1

2.3 By Age Group tab

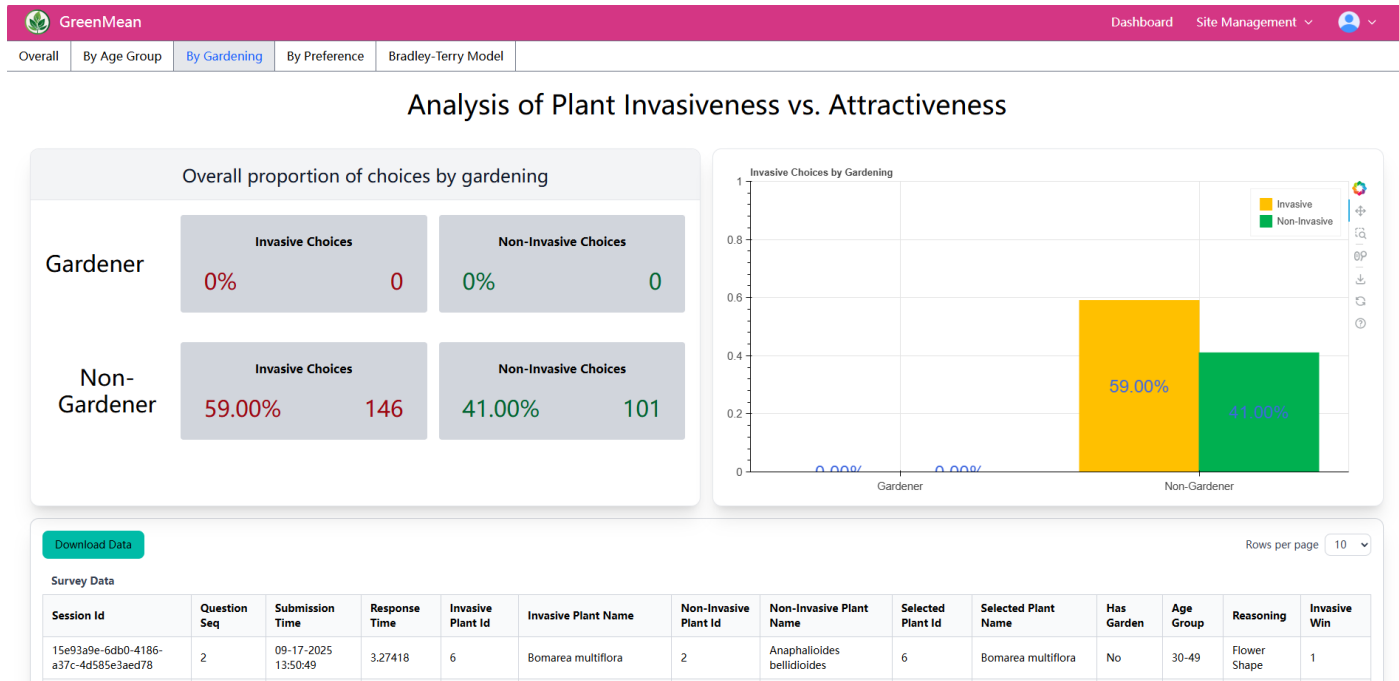
Similarly, the top section displays summary numbers and a bar chart for each age group. Below, a table shows the raw survey data, which can be downloaded.

Analysis of Plant Invasiveness vs. Attractiveness



2.4 By Gardening tab

Similarly, the top section displays summary numbers and a bar chart grouped by gardener and non-gardener. Below, a table shows the raw survey data, which can be downloaded.



2.5 By Preference tab

At the top, the page displays the overall percentage and count for each preference chosen by survey participant. For each preference, it also shows the percentage and count broken down by age group and gardener/non-gardener. At the bottom, a table presents the raw survey data, which can be downloaded.

Survey Preference Summary

Preference	Overall		18-29		30-49		50-64		65+		Gardener		Non-Gardener	
	%	Count	%	Count	%	Count	%	Count	%	Count	%	Count	%	Count
Dominant Flower Colour	8.00%	2	50.00%	1	50.00%	1	0.00%	0	0.00%	0	0.00%	0	100.00%	2
Contrasting Colours	32.00%	8	12.50%	1	87.50%	7	0.00%	0	0.00%	0	0.00%	0	100.00%	8
Flower Shape	60.00%	15	0.00%	0	100.00%	15	0.00%	0	0.00%	0	0.00%	0	100.00%	15
Familiarity	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0
Different From Most Plant You Know	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0	0.00%	0

[Download Data](#)

Rows per page10

Session Id	Submission Time	Has Garden	Age Group	Preference
15e93a9e-6db0-4186-a37c-4d585e3aed78	09-17-2025 13:50:39	0	30-49	Flower Shape
89cd182d-7b05-47b7-ab5a-69b6b62b2ca4	09-17-2025 20:45:36	0	30-49	Flower Shape
09f8ef2a-32a2-4634-8ff7-bbcc86d72f5e	09-17-2025 21:15:53	0	18-29	Dominant Flower Colour
8c322364-7733-4fb5-a1e3-c42ff17f0966	09-17-2025 22:41:24	0	30-49	Flower Shape
5a65e8cc-1bfa-4aab-85bb-dd47b744bc8f	09-17-2025 22:51:32	0	30-49	Flower Shape
9fcb1332-02f1-496d-9a5b-91f7f9cf07b9	09-17-2025 22:57:41	0	30-49	Flower Shape

2.6 Bradley-Terry Model tab

This page displays visualizations of 6 charts which are based on Bradley-Terry model. Below each chart, a table shows the corresponding analysis data, which can be downloaded.

2.6.1 Bradley-Terry Beta vs Win Percentage

This chart displays the normalized Bradley-Terry beta score versus the win percentage for each plant. The left chart displays the unweighted version, while the right chart shows the weighted version (taking into account the response time for each selection).

Invasive plants are marked in yellow, and non-invasive plants are marked as green.

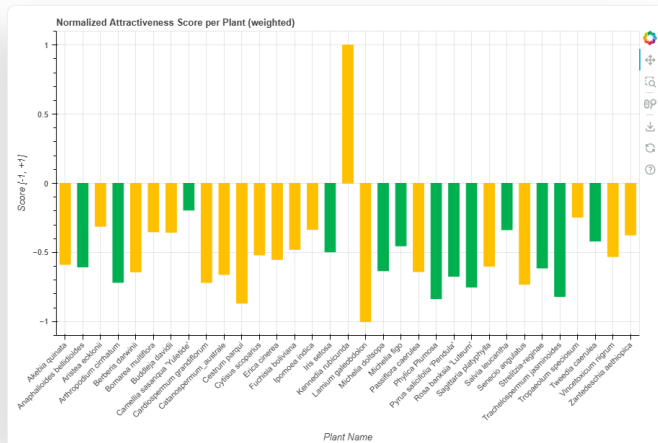
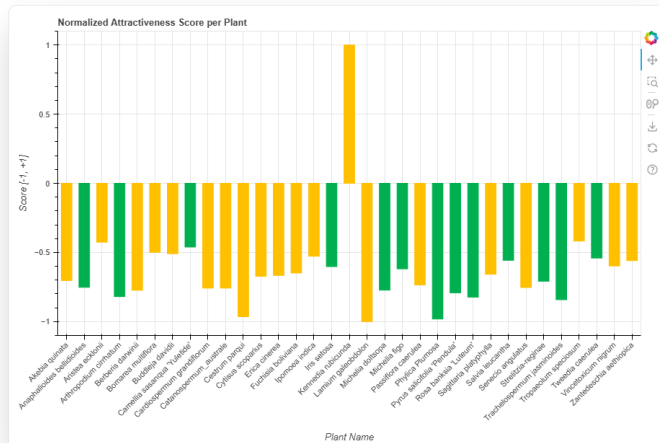
Bradley-Terry Beta vs Win Percentage



2.6.2 Bradley-Terry Beta by each plant

This chart shows the normalized Bradley-Terry beta score for each plant. The left chart displays the unweighted version, while the right chart shows the weighted version. Invasive plants are marked as yellow, non-invasive plants are marked as green.

Bradley-Terry Beta By Each Plant



2.6.3 Ranking of Bradley-Terry Beta by Each Plant

This chart displays the normalized Bradley-Terry beta score for each plant, sorted by their ranking.

The left chart displays the unweighted version, while the right chart shows the weighted version.

Invasive plants are marked as yellow, non-invasive plants are marked as green.

Ranking of Bradley-Terry Beta By Each Plant

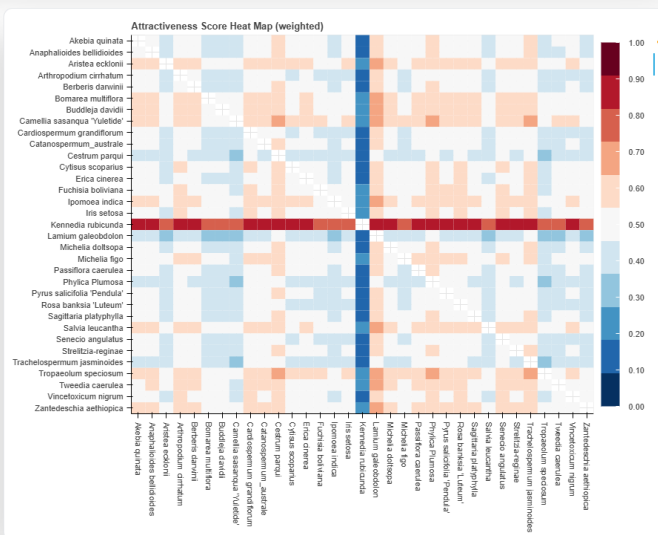
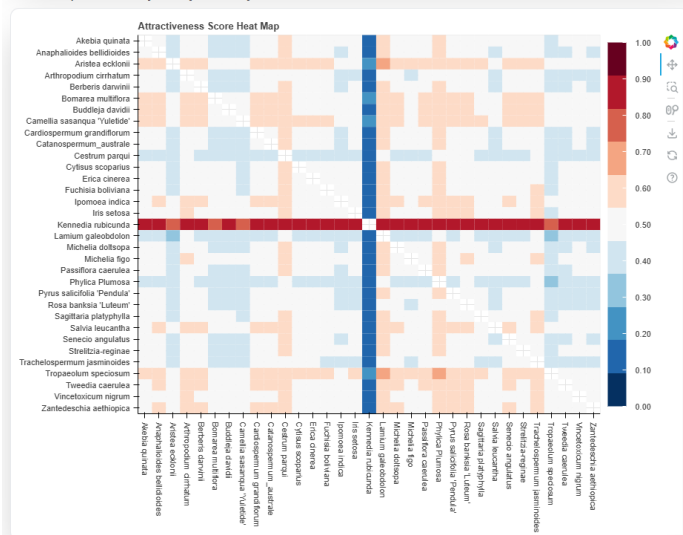


2.6.4 Heat map of Bradley-Terry Beta by Each Plant

This chart displays a heat map of the probability for each plant wins against another plant.

The left chart shows the unweighted version, while the right chart displays the weighted version.

Heat map of Bradley-Terry Beta by Each Plant



2.6.5 Histogram of Bradley-Terry Beta by Invasive Type

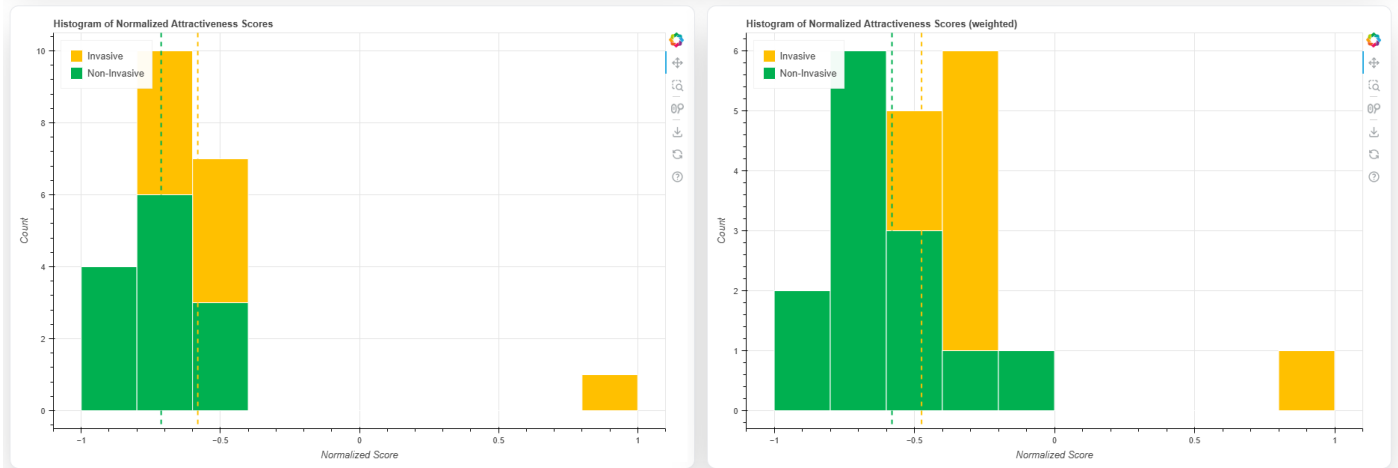
This chart shows the histogram of normalized Bradley-Terry beta score grouped by invasive or non-invasive type.

The left chart is the unweighted version, while the right chart is the weighted version.

Invasive plants are marked as yellow, and non-invasive plants are marked as green.

The two dotted lines indicate the mean beta scores.

Histogram of Bradley-Terry Beta By Invasive Type



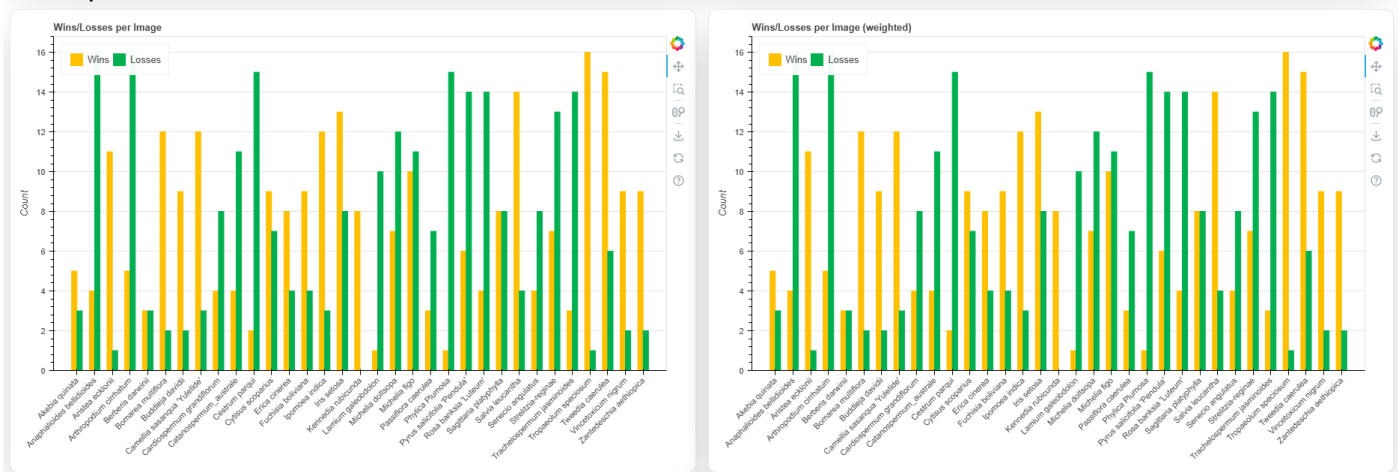
2.6.6 Win/Loss by each plant

This chart shows the win and loss count for each plant.

The left chart displays the unweighted version, and the right chart shows the weighted version. However, since response time does not affect win/loss counts, both versions are the same.

Invasive plants are marked as yellow, and non-invasive plant are marked as green.

Win/Loss By Each Plant



2.6.7 Quick navigation

Click the hovered blue navigation button  to quick navigate between the six charts.

Charts Navigation

×

1. Bradley-Terry Beta vs Win Percentage

2. Bradley-Terry Beta By Plant

3. Bradley-Terry Beta By Plant (Ranking)

4. Bradley-Terry Beta By Plant (Heatmap)

5. Bradley-Terry Beta By Invasive Type (Histogram)

6. Win/Loss By Plant

2.6.8 Add new plant image

Go to Site Management -> Add Plant

Users can add a new plant by uploading an image, entering name and description, and specify whether the plant is invasive or non-invasive.

Add Plant



Upload Image

Choose File 屏幕截图 2025-08-07 103045.png

Name

Rosa

Description

the description for Rosa

Invasiveness

Invasive

Add Plant

Cancel

2.6.9 Plant List & Edit

Click Site Management -> Plant List

A list of plants is displayed. To modify a plant or its image, click the edit or delete button.

Plant List				
Add a new Plant				
NAME	DESCRIPTION	INVASIVENESS	IMAGE	ACTIONS
Zantedeschia aethiopica	Zantedeschia aethiopica Weed	invasive		Edit Delete
Vincetoxicum nigrum	Vincetoxicum nigrum Weed	invasive		Edit Delete
Tweedia caerulea	Tweedia caerulea NotWeed	non-invasive		Edit Delete
Tropaeolum speciosum	Tropaeolum speciosum Weed	invasive		Edit Delete
Trachelospermum jasminoides	Trachelospermum jasminoides NotWeed	non-invasive		Edit Delete
Strelitzia-reginae	Strelitzia-reginae NotWeed	non-invasive		Edit Delete

2.6.10 Survey Cycle Management

Click Site Management -> Survey Cycle

Survey Summary				
TOTAL SURVEY PARTICIPANTS	TOTAL CHOICES	SURVEY DATA	ACTION	
25	247	Download	Update	

Survey Cycle							
Rows per page: 10							
INTERNAL CYCLE ID	START TIME	END TIME	SURVEY PARTICIPANTS	TOTAL CHOICES	STATUS	SURVEY DATA	ACTION
1	17-09-2025 13:50:24	17-09-2025 13:51:47	1	9	Closed	Download	Update Delete
2	17-09-2025 13:51:47	27-09-2025 11:15:56	10	98	Closed	Download	Update Delete
4	27-09-2025 11:15:56	04-10-2025 02:28:36	5	50	Closed	Download	Update Delete
5	04-10-2025 02:28:36		9	90	Active	Download	Update Close Delete

Showing 1 to 4 of 4 entries

Prev Page 1 of 1 Next

At the top of the page, the survey summary for all survey results is displayed, and the raw data is downloadable. Click the update button to refresh the analysis data if any raw data is changed.

At the bottom of the page, summaries for each survey cycle are shown. The current survey cycle is active, while others are closed. Click the close button to close current active cycle and start a new survey cycle.

Each survey cycle also has a **Download** button to export the cycle's data, and a Delete button to remove the whole cycle.

Click on a survey cycle row, to view a detailed table showing the raw survey data within that cycle. Admin users can delete individual entries clicking the delete icon. This allows the removal of invalid data (e.g. unusually too long & abnormal response times).

After deleting the survey raw data, click the **Update** button for the survey cycle to refresh its analysis data. Also, click the **Update** button for the overall survey summary.

Survey Cycle							
INTERNAL CYCLE ID	START TIME	END TIME	SURVEY PARTICIPANTS	TOTAL CHOICES	STATUS	SURVEY DATA	ACTION
1	17-09-2025 13:50:24	17-09-2025 13:51:47	1	9	Closed		Update
2	17-09-2025 13:51:47	27-09-2025 11:15:56	10	98	Closed		Update
4	27-09-2025 11:15:56	04-10-2025 02:28:36	5	50	Closed		Update
5	04-10-2025 02:28:36		9	90	Active		Update Close

Showing 1 to 4 of 4 entries

Prev Page 1 of 1 Next

Cycle Details

SESSION ID	QUES. SEQ	SUBMITTED AT	RESPONSE TIME	INVASIVE PLANT NAME	NON-INVASIVE PLANT NAME	SELECTED PLANT NAME	
15e93a9e-6db0-4186-a37c-4d585e3aed78	2	09-17-2025 13:50:49	3.27418	Bomarea multiflora	Anaphalioides bellidioides	Bomarea multiflora	
15e93a9e-6db0-4186-a37c-4d585e3aed78	3	09-17-2025 13:50:52	2.82467	Cestrum parqui	Michelia doltosopa	Michelia doltosopa	
15e93a9e-6db0-4186-a37c-4d585e3aed78	4	09-17-2025 13:50:56	3.06776	Sagittaria platyphylla	Iris setosa	Iris setosa	
15e93a9e-6db0-4186-a37c-4d585e3aed78	5	09-17-2025 13:50:59	2.7806	Catanospermum australe	Tweedia caerulea	Tweedia caerulea	

2.6.11 Profile management

Click the user icon on the top right and select My Profile. Admin users can edit profile information.

User Profile

User Details

Username	siteadmin1
Email	siteadmin1@example.com
First Name	John
Last Name	Doe
Description	
Site Role	siteadmin

Location

Update Profile

2.6.13 Bradley-Terry Model

For calculation of Bradley-Terry model, the following formulas are used.

Probability that i beats j : $P(i \text{ beats } j) = \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}}$

Likelihood function: $l(\beta) = \sum w_{ij} \cdot \log P(i \text{ beats } j) = \sum w_{ij} \cdot \log \frac{e^{\beta_i}}{e^{\beta_i} + e^{\beta_j}}$

Weight(inverse weighting): $w_{ij} = \frac{1}{RT_{ij} + 10^{-9}}$

RT: Response Time

For un-weighted model, weight = 1.

To calculate the beta value for each plant, the **minimize** function from python package scipy.optimize library is used.